

Math

Section - A

1. HCF of 36 and 45

$$36 = 2^2 \times 3^2 \times 1$$

$$45 = 3^2 \times 5$$

(3) 3

$$2. \frac{2 \tan 30^\circ}{1 - \tan^2 30^\circ}$$

$$2 \left(\frac{1}{\sqrt{3}} \right)$$

$$\frac{\sqrt{3}}{1 - \left(\frac{1}{\sqrt{3}} \right)^2}$$

$$= \frac{2}{1 - \frac{1}{3}} = \frac{2}{\frac{2}{3}} = 2 \times \frac{3}{2} = 3$$

$$\frac{2 \times 3 \sqrt{3}}{\sqrt{3} \times 2}$$

$$= \sqrt{3}$$

$$(3) \tan 60^\circ$$

$$3. \frac{PA}{PD} = \frac{6}{5}$$

$$\frac{PB}{PC} = \frac{3}{2.5} = \frac{6}{5}$$

$$\therefore \frac{PA}{PD} = \frac{PB}{PC}$$

$$\angle APB = \angle DPC \text{ (V.O.A.)}$$

By SAS

$$\triangle APB \sim \triangle DPC$$

$$\angle A = \angle D = 30^\circ$$

$$\angle A + \angle P + \angle B = 180^\circ$$

$$30^\circ + 50^\circ + \angle B = 180^\circ$$

$$\angle B + \angle C = 180^\circ$$

$$B = 100$$

$$K_{100}$$

4. ~~max and min~~

$$(D) 100$$

$$5. \quad x + y \Rightarrow 2 + (-2)$$

$$= -1$$

$$x \times y$$

$$= 2 \times -2$$

$$= -4$$

$$-b = -1$$

$$-(a+1) = -1$$

$$-a-1 = -1$$

$$-a = -1 + 1$$

$$-a = 0$$

$$a = 0$$

$$(1) a = 0$$

6. KD four decimal places

$$7. \begin{aligned} a-y &= 2 \\ a-b &= 2 \\ a+b &= 4 \end{aligned}$$

Add them

$$\begin{aligned} a+b &= 2 \\ a+b &= 4 \\ \hline 2a &= 6 \\ a &= 3 \end{aligned}$$

$$3-b=2$$

$$-b=-1$$

$$b=1$$

(3) 3 and 1

$$8. \left(\frac{8+(-2)}{2}, \frac{3+3}{2} \right) = \left(\frac{6+x}{2}, \frac{7+y}{2} \right)$$

$$\left(\frac{6-2}{2}, \frac{6}{2} \right) = \left(\frac{6+x}{2}, \frac{7+y}{2} \right)$$

$$\left(\frac{6}{2}, \frac{6}{2}\right) = \frac{6+x}{2}, \frac{3+y}{2}$$

$$(3, 3) = \frac{6+x}{2}$$

$$6+x=6$$

$$x=0$$

$$6+x+y=6$$

$$y=-1$$

$$(0, -1)$$

$$\Rightarrow (2)(0, -1)$$

$$9. \frac{a_1}{a_2} = \frac{9}{18} = \frac{1}{2}$$

$$\frac{b_1}{b_2} = \frac{3}{6} = \frac{1}{2}$$

$$\frac{c_1}{c_2} = \frac{12}{26} = \frac{6}{13} \neq \frac{1}{2}$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

§(10) No solution

10.

$$\frac{a_1}{a_2} = \frac{b_1}{b_2}$$

$$x + 2 = (9, 18)$$

$$2 = x + 2$$

$$0 = x$$

$$\frac{2}{3} = \frac{-3}{k}$$

$$2 = -3k$$

$$1 = -\frac{3}{2}k$$

$$k = -\frac{2}{3}$$

$$-2k = -9$$

$$k = \frac{9}{2}$$

$$\frac{2}{3} = \frac{k-0}{(1-0)(1)} \in$$

$$1 = \frac{k}{1} = k$$

$$k = 1$$

$$\frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\frac{2}{3} = \frac{2k}{1}$$

$$k = \frac{3}{2}$$

$$1 = \frac{3}{2} \cdot \frac{2}{1} = 3$$

$$1 = 3$$

$$\frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\frac{2}{3} = \frac{-3}{k}$$

$$1 = \frac{-3}{k} \cdot k = -3$$

$$1 = -3$$

$$2k = -9$$

$$k = \frac{-9}{2}$$

$$\Rightarrow (1) k = -9$$

$$11 \quad \frac{a_1 \neq b_1}{a_2 \neq b_2}$$

$$\frac{10 \neq -5}{6 \neq k}$$

$$10k \neq -30$$

$$k \neq -3$$

$$\Rightarrow (1) k \neq -3$$

$$12 \quad \text{HCF of } (32, 616)$$

$$32 = 2^5$$

$$616 = 2^3 \times 19$$

$$\text{HCF} = 2^3$$

$$= 8$$

$$\Rightarrow (1) 8$$

13. (1) similar

$$\begin{array}{r} 2 \overline{) 616} \\ 4 \\ \hline \end{array}$$

$$\begin{array}{r} 2 \overline{) 38} \\ 19 \\ \hline \end{array}$$

$$\begin{array}{r} 19 \overline{) 19} \\ 1 \\ \hline \end{array}$$

14. $a = x^3 y^4 z^2$
 $b = x^2 y^4 z^4$

H $\text{LCM}(a, b) = x^3 y^4 z^4$

(1) $x^3 y^4 z^4$

15. $\alpha + \beta = -3 + 4 = 1$

$\alpha \times \beta = 4 \times -3 = -12$

$x^2 - 1x - 12$

16. (3) Intersecting or coincident

17. (1) Parallel

18. $\frac{\text{ar}(\Delta ABC)}{\text{ar}(\Delta PQR)} = \left(\frac{4}{9}\right)^2$

$\frac{\text{ar}(\Delta ABC)}{\text{ar}(\Delta PQR)} = \frac{3 \times 2}{3}$

$$\Rightarrow 11) 2:3$$

$$19. x^2 - 4x = 5$$

$$x^2 - 4x + (x - 5)$$

$$x(x - 5) + 1(x - 5)$$

$$(x + 1)(x - 5)$$

$$x = -1 \text{ or } x = 5$$

$$x + b = 2 + \sqrt{3} + 2 - \sqrt{3}$$

$$= 4$$

$$x + b = 4 - 3$$

$$= 1$$

$$x^2 - 4x + 1$$

(3) A is true but R is false

$$20. \frac{4}{3(4+1)} = \frac{20}{15}$$

$$\frac{20}{15} = \frac{4}{3}$$

$$\frac{20}{15} = \frac{4}{3}$$

$$4(3k-2) = 20 \times 15$$

$$12k - 8 = 300$$

$$12k = 308$$

$$k = \frac{308}{12}$$

$$k = 26$$

(4) A is false but

R is true

$$21 \quad 5(6 \times 11 \times 3 \times 2 \times 1 + 1)$$

$$5(75)$$

$$5(145)$$

It means that 5 and

145 are the factors of

$$6 \times 11 \times 3 \times 2 \times 1$$

$$6 \times 11 \times 3 \times 2 \times 1 + 1$$

$$6 \times 11 \times 3 \times 2 \times 1 + 1$$

is a composite number

$$22. \frac{y}{k} = \frac{6}{16}$$

$$\frac{3 \cancel{18}}{\cancel{X} 4}$$

$$x_2 = 6k$$

$$k = 12$$

$$23. \cancel{x_2 + 12}$$

$$\sqrt{x^2 + y^2}$$

$$= \sqrt{(-6)^2 + (8)^2}$$

$$= \sqrt{36 + 64}$$

$$= \sqrt{100}$$

$$\Rightarrow 10 \text{ units}$$

$$24. \cos(90^\circ - 30^\circ) = \cos(A - 16^\circ)$$

$$90 - 3A = A - 16$$

$$90 + 16 = 4A$$

$$106 = 4A$$

$$A = 26.5$$

25. AB11CD

$$\angle AOB = \angle COD \text{ (V.O.A.)}$$

$$\angle OAB = \angle ODC \text{ (alt int angles)}$$

$$\angle OCD = \angle OBA \text{ (alt int angles)}$$

By AAA similarity

$$\Delta AOB \sim \Delta DOA$$

26. LHS

$$\sin(90^\circ - (40^\circ - \theta)) - \cos(40^\circ - \theta) + \tan(30^\circ - 89^\circ) + \tan(90^\circ - 80^\circ)$$

$$= \sin(50^\circ + \theta) - \cos(40^\circ - \theta)$$

$$= (\tan 30^\circ + \tan 80^\circ \tan 90^\circ)$$

$$\Rightarrow (\sin 40^\circ - \theta) \cos(40^\circ - \theta) - \cos(40^\circ - \theta)$$

$$+ \cos 40^\circ \cos 80^\circ \cos 70^\circ \tan 70^\circ$$

$$= \tan 80^\circ \tan 90^\circ$$

$$\Rightarrow \cos 89^\circ \cos 80^\circ \cos 70^\circ + \tan 80^\circ \tan 90^\circ$$

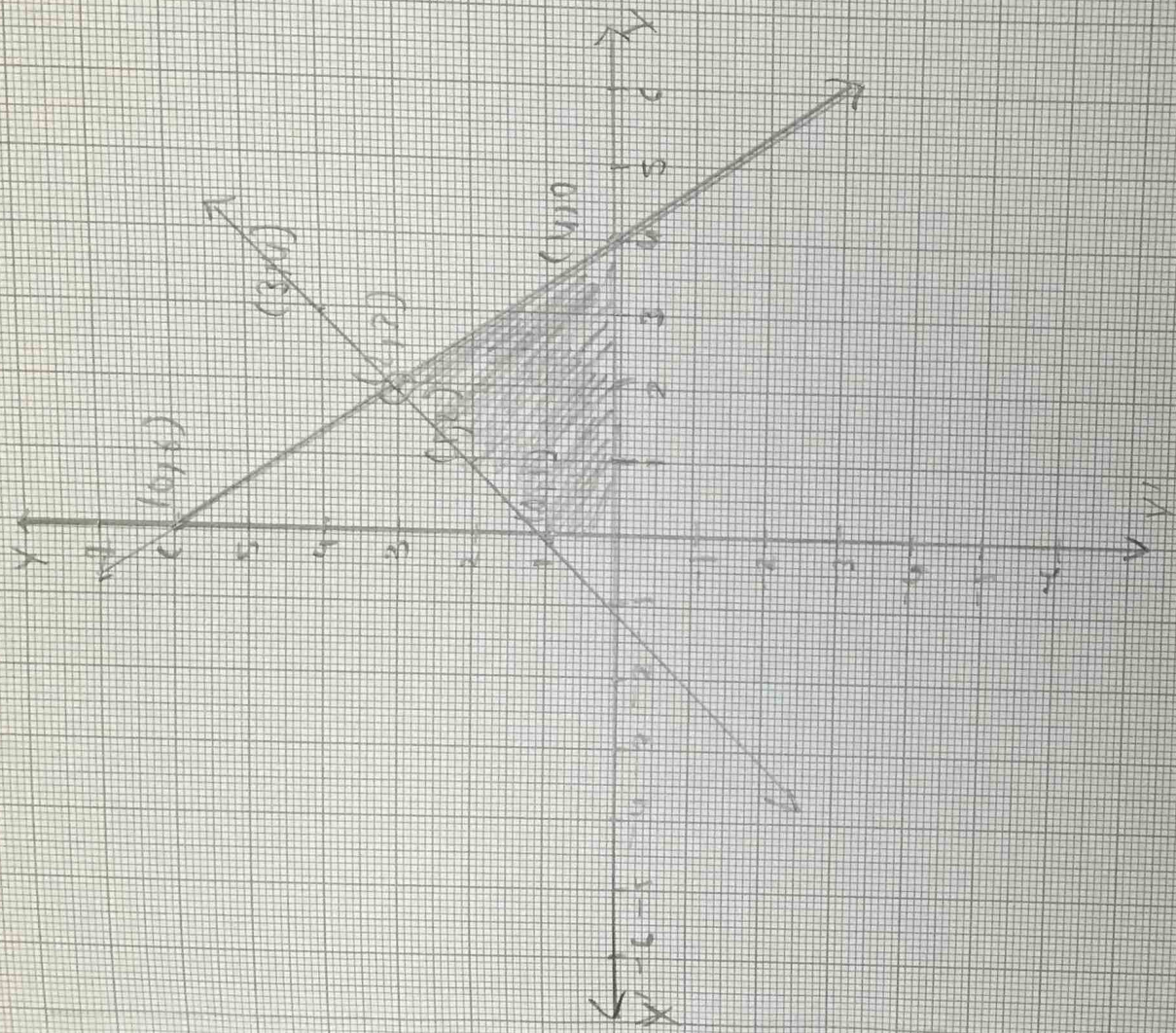
$$\cos 0^\circ = 1$$

$$\tan 0^\circ$$

$$\Rightarrow \tan 70^\circ + \tan 80^\circ + \tan 90^\circ$$

$$= \tan 70^\circ + \tan 80^\circ + \tan 90^\circ$$

$$\Rightarrow 1 = RHS$$



$$35 \quad p(x, y) = (h(x) + m_1 + m_2) \cdot (m_1 y_1 + m_2 y_2 + m_1 + m_2)$$

$$\Rightarrow \frac{3(8) + (8)5}{3+1}, \frac{3(5) + 1(-3)}{3+1}$$

$$x = \frac{h + n_2}{4}$$

$$= \frac{28}{4}$$

$$x = 7$$

$$y = \frac{3(5) + 1(-3)}{3+1}$$

$$y = \frac{15 - 3}{3+1}$$

$$y = \frac{12}{4}$$

$$y = 3$$

$$p(x, y) = (7, 3)$$

$$\text{solution} = (2, 3)$$

$$x=2 \text{ and } y=3$$

$$3x - 2y = 12$$

$$3 \times 2 - 0 \times 3 = 12$$

$$\text{Area} = \frac{1}{2} \times B \times h$$

$$\frac{1}{2} \times 5 \times 3$$

$$\frac{1}{2} \times 15$$

$$= 7.5 \text{ sq units}$$

$$3x^2 - 2x - 3 = 12$$

$$3x^2 - 2x - 15 = 12$$

$$3x^2 - 2x - 15 = 12$$

$$34. \quad 2x^3 - x^2 - 4x + 2$$

Let α, β, γ be zeroes of the polynomial

$$\alpha + \beta + \gamma = \frac{-b}{a}$$

$$\sqrt{2} + (-\sqrt{2}) + \gamma \Rightarrow \frac{-(-1)}{2}$$

$$0 + \gamma \Rightarrow \frac{1}{2}$$

$$\Rightarrow \gamma \Rightarrow \frac{1}{2}$$

26

(i) (2) 10, 12

alternative (3) 13

(ii) (1) 3, 5

$\Sigma =$ because we know values (11)

$$1 - \Sigma = (5 - 3) + 5 = 7 + 5$$

$$2 - \Sigma = 5 - 3 - 5 = -3$$

(iii) P(3, 5) Q(6, 9)

$$d = 10/4 + 5/10$$

$$\sqrt{(6-3)^2 + (9-5)^2}$$

$$1 - \Sigma = 5 - 3 + 5 (5 -)$$

$$= \sqrt{(3)^2 + (4)^2}$$

$$2 - \Sigma = 5 - 3 - 5$$

$$1 - \Sigma = 5 - 3 - 5$$

$$= \sqrt{9 + 16}$$

$$= \sqrt{25}$$

$$= 5 \text{ units}$$

(3) 5m

$$1, 2 - (3) = 110$$

~~(1)~~

$$1, 2 - (3) = 110$$

$$1, 2 - (3) = 110$$

$$1, 2 - (3) = 110$$

37. (3) parabola

iii) Zeros are 2 and -3
 $\alpha + \beta = 2 + (-3) = -1$
 $\alpha \times \beta = 2 \times -3 = -6$

$$x^2 + 1x - 6$$

$$(-3)^2 + 1x - 3 - 6$$

$$9 - 3 - 6 = 0$$

(3) 0

(ii) (3) -3, 2

38.

(i) (3) 2, 4, 1

(ii) (1) R.H.S

(iii) (2) 4, 9